



more assistance. The team's research demonstrated that it can acquire a new task in as little as 25 minutes. This model opens up possibilities for robots to venture into the surrounding world with a sense of curiosity. To guide the robot in understanding object interaction, the team incorporated the idea of affordances, drawing from psychological principles. In the virtual robotic behaviour (VRB) context, affordances serve as guidelines for determining the location and manner in which a robot can interact with an object, drawing insights from human behaviour. The researchers believe that this research can help robots learn from online videos.

Can synthetic fuel replace fossil fuel?

Porsche, in collaboration with HIF Global, Siemens Energy, ExxonMobil and other international partners, has started the industrial production of synthetic fuels. Porsche's eFuel is a sustainable alternative to gasoline, made with renewable electricity, water, and carbon dioxide. It reduces emissions and showcases Porsche's commitment to sustainable mobility. This innovative approach lowers carbon emissions by using renewable electricity for production. eFuel's carbon emissions are limited to the initial carbon dioxide used, making it environmentally responsible. It seamlessly fits existing infrastructure and can be pumped at standard stations without vehicle modifications. This smooth transition allows drivers to adopt eFuel easily without major changes or investments. Porsche plans to use eFuel in hybrid/electric cars to extend range and reduce the carbon footprint.

Perovskite solar cells with 24.35% efficiency

The National University of Singapore (NUS) researchers have achieved a groundbreaking efficiency of 24.35% in an inverted-structure perovskite solar cell, setting a new record for the highest ef-



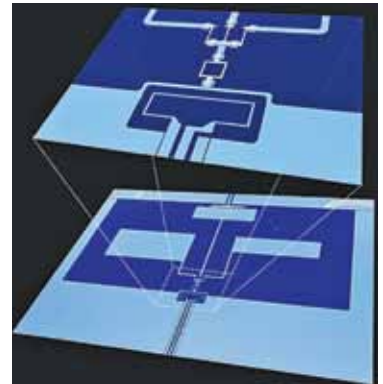
Perovskite solar cells (Credit: <https://www.mercomindia.com>)

iciency achieved in a perovskite solar cell with a 1-centimetre square active area. The researchers have achieved higher efficiency in perovskite solar cells by incorporating a novel interface material with synergistic properties. This improves performance and durability, making perovskite cells more commercially feasible. To enhance long-term viability, the team is actively working on stabilising perovskite materials, which are prone to degradation in moisture.

Their primary objective is developing a tailored ageing methodology to ensure 25 years of operational stability. This advancement holds promise for the future of solar energy, offering a more efficient and durable alternative to conventional perovskite cells, ultimately contributing to the transition to sustainable and renewable power sources.

NIST's toggle switch can help quantum computers cut through noise

Scientists at the National Institute of Standards and Technology (NIST) have created a device with two superconducting qubits, quantum counterparts of classical computer bits. A 'toggle switch' connects the qubits to a 'readout resonator' circuit. The researchers aim to maintain undisturbed yet read-



Central working region of the device (Credit: K. Cicak and R. Simmonds/NIST)

able outputs, safeguarding qubits and enhancing measurement fidelity for quantum processors. Quantum computers exploit quantum mechanics for complex tasks like drug development. Designers face noise issues affecting quantum circuits. The toggle switch overcomes problems by blocking circuit noise and preventing undesired qubit interactions. Remote-controlled microwave pulses replace physical connections in a static architecture, enabling a programmable quantum computer. More toggle switches enhance versatility while facilitating logic operation order and simultaneous qubit measurement. The toggle switch, made from a sensitive SQUID device, interacts with qubits and the readout resonator via microwave current.

A sensor chip 10x more sensitive at disease diagnostics

Texas A&M AgriLife Research scientists and collaborators at Iowa State University have developed a sensor chip that can detect many disease pathogens, at a level ten times greater than existing detection methods. The new sensor enhances pathogen detection through loop-mediated isothermal amplification (LAMP) by eliminating expensive fluorescence dye labelling. It detects pathogens without reagents, offers high sensitivity, and eliminates the need for time-consuming DNA purification. The sensor includes